

Experimental Results Summary

DPA-VMI (Dual-Perspective Variance-Tuned Attack) - Face Verification

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Experimental Setup

Attack: DPA-VMI (Dual-Perspective + VMI)

Papers: CVPR 2021 & CVPR 2025

Integration: Shared transfer attack pipeline

Evaluation Metrics:

- Breach Success Rate (%)
- Embedding Impact (Mean Cosine Distance Shift)

Key Observations

- Combines 3D perspective views (DPA) with neighborhood gradient variance tuning (VMI), Nesterov (NI), and Gaussian blur (TI).
- Achieves **Rank #1 overall** across all evaluated baseline attacks on 480 test cases.
- Reaches **37.92% overall breach rate** (+8.75 pp higher than SI-NI-FGSM).
- Induces a high **0.2101 mean impact**.

Overall Performance

Attack Method	Breach Rate (%)	Mean Impact
DPA_VMI (Proposed)	37.92	0.2101
SI-NI-FGSM	29.17	0.1755
MI-FGSM	26.67	0.1643
MI-ADMIX-DI-TI	24.17	0.1534
TI-FGSM	20.42	0.1313
PGD	16.67	0.1002

Attack Goal Breakdown

Goal	Breach Rate (%)	Mean Impact
Dodging	47.08	0.2652
Impersonation	28.75	0.1549

Conclusion

Dual-Perspective Transformation and Variance Tuning can be effectively integrated for face verification transfer attacks. By combining 3D perspective geometric warps (DPA) with neighborhood gradient variance correction (VMI), Nesterov lookahead (NI), and Gaussian depthwise convolution (TI), DPA_VMI eliminates surrogate-specific gradient overfitting. Evaluated on 480 non-self transfer cases across 5 CNN face architectures, DPA_VMI achieved an overall breach rate of 37.92% and a mean impact of 0.2101, outperforming SI-NI-FGSM by +8.75% and PGD by +21.25%.